

Lattice Awakening Cosmology

*v5.0 — Void-Driven Coasting Expansion,
Geometric Suppression, and the Emergent Hubble Parameter*

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Abstract

We present Lattice Awakening Cosmology v5.0, a discrete phenomenological model in which spacetime emerges from the sequential activation of fundamental lattice units of scale $l_L \approx l_P$. Cosmic expansion is not a primary force but the macroscopic consequence of a single local stability rule: every activated lattice site must possess $N_s = 6$ active face-adjacent neighbours (simple-cubic geometry); any deficit triggers the activation of the missing neighbour.

The quantum support state ($S = 0$) exerts a resistance force exactly equal and opposite to the total gravitational force applied to it ($F_{\text{sup}} = -F_{\text{grav}}$). Direct substitution into the boundary equation of motion yields $\ddot{r} = 0$: the universe expands at *constant velocity*. From $r(t) = \dot{r} t$ we obtain

$$H(t) = \frac{1}{t}, \quad H_0 = \frac{1}{t_0} \approx 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

with no free parameters. This value lies between the two branches of the Hubble tension (67.4 and 73.0 $\text{km s}^{-1} \text{ Mpc}^{-1}$); the tension itself is reinterpreted as a lattice rendering-resolution effect. The cosmological constant Λ is not fundamental: it is replaced by a derived void-density term ρ_v that emerges from the surface-to-volume ratio of the activation front. The Friedmann equation is derived from first principles.

1. Foundational Definitions, Axioms, and Equations

This section collects *all* primitive definitions, axioms, and derived equations that the rest of the paper depends upon. Nothing is introduced later without reference to an entry here.

1.1. Primitive Definitions

Definition 1 (Lattice scale).

$$l_L \approx l_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m}.$$

The size of a single lattice site is invariant. l_L is the Planck length as a candidate value; it is treated as the primary free parameter of the model until confirmed by the Hubble calculation in Section 5.

Definition 2 (Lattice site and state). Each point $\mathbf{x} \in \mathbb{Z}^3$ carries a binary state

$$S(\mathbf{x}, t) \in \{0, 1\}.$$

$S = 1$: **activated** — the site is a real spacetime coordinate; gravity $= G_L$ is generated; physical laws apply.

$S = 0$: **quantum support state** — the site is in the ICP Transmission phase (timeless, spaceless, gravityless).

Definition 3 (Neighbour set — simple-cubic).

$$\mathcal{N}(\mathbf{x}) = \{\mathbf{x}' \in \mathbb{Z}^3 : |\mathbf{x} - \mathbf{x}'|_1 = l_L\}, \quad |\mathcal{N}(\mathbf{x})| = N_s = 6.$$

The ℓ^1 norm selects the six face-adjacent sites of the simple-cubic lattice.

Definition 4 (Active-neighbour count).

$$N_{\text{neighbors}}(\mathbf{x}, t) = \sum_{\mathbf{x}' \in \mathcal{N}(\mathbf{x})} S(\mathbf{x}', t).$$

Definition 5 (Lattice density).

$$\rho_L(\mathbf{x}, t) = \frac{N_{\text{neighbors}}(\mathbf{x}, t)}{N_s l_L^3}.$$

Mass density is identified with lattice density: no separate mass concept is required.

1.2. Axioms

Axiom 1 (Quantum pre-computation phase). Before the first activation event there exists a timeless, spaceless, gravityless state—the ICP *Transmission* phase—in which all physical laws are computed as probability distributions. $\Delta t = 0$. Quantum non-locality follows as a formal consequence: correlations require no signal-transit time because no time exists.

Axiom 2 (Quantum instantaneity). In the Transmission phase $\Delta t = 0$. All quantum computations, including the complete lattice-placement map, are finalised before the first activation.

Axiom 3 (The Big Bang as first activation). The Big Bang is the activation of a single lattice site. It has a centre. Apparent isotropy arises from the uniform activation rule applied to a spherically expanding front, not from the absence of a centre. Inflation is unnecessary: the homogeneity problem does not arise because the activation rule is identical at every point on the front.

Axiom 4 (Spacetime = gravity = lattice). Each activated lattice site *is* gravity; it is not a substrate on which gravity acts. The gravitational field is the gradient of the lattice potential, not the density itself. A uniform lattice produces zero net gravitational field.

Axiom 5 (Stability condition — core dynamical rule).

$$N_{\text{neighbors}}(\mathbf{x}, t) \geq N_s \implies \text{stable},$$

$$N_{\text{neighbors}}(\mathbf{x}, t) < N_s \implies S(\mathbf{x}, t + \Delta t) = 1 \quad (\text{activation triggered}).$$

A deficit of active neighbours is a *void*. Activation is void-filling. Cosmic expansion is the macroscopic manifestation of this rule.

Axiom 6 (Support-state resistance — energy minimisation). The quantum support state ($S = 0$) resists activation with a force exactly equal and opposite to the total gravitational force applied to it:

$$F_{\text{sup}} = -F_{\text{grav}}.$$

This is not an ad-hoc assignment. It follows from the energy-minimisation condition

$$\frac{\delta E_{\text{total}}}{\delta S(\mathbf{x}, t)} = 0 \iff N_{\text{neighbors}}(\mathbf{x}, t) \geq N_s,$$

where

$$E_{\text{total}} = E_{\text{grav}} + E_{\text{sup}} = -\frac{1}{2}G_L\rho_L\Phi + \frac{1}{2}G_L\rho_L\Phi = 0.$$

Axiom 7 (Lattice expansion = new site generation). l_L is invariant. Cosmic expansion is not the stretching of existing sites but the activation of previously dormant $S = 0$ sites. Matter-internal lattice sites are unaffected; atomic scales do not change with cosmic expansion.

Axiom 8 (Black holes as lattice superposition). A black-hole singularity is not infinite density. It is maximal lattice superposition: multiple activated lattice layers occupying the same coordinate. Information (lattice state) is conserved.

Axiom 9 (Universe as black-hole interior — Digital Unzipping). A black hole containing the total mass-energy of our universe would, from its interior, appear as a stable, isotropic, lattice-filled universe. The 360° isotropy of lattice inter-site forces explains large-scale homogeneity.

Crucially, the early universe does not collapse back into a singularity because it is not an analog explosion, but a ‘**Digital Unzipping**’ process. Quantum information from the support state is sequentially realized into the spacetime grid ($H = 1/t$). The high initial density is the ‘input pressure’ of information, and the expansion is the ‘output rendering’ of the system.

1.3. Core Equations (Summary)

Label	Equation	Source
Lattice Poisson	$\nabla^2 \Phi = 4\pi G_L \rho_L$	Axiom 4
Gravity field	$\mathbf{g} = -\nabla \Phi$	Axiom 4
Support force	$F_{\text{sup}} = -F_{\text{grav}}$	Axiom 6
Activation front speed	$\dot{r} = c$	Section 3
Boundary acceleration	$\ddot{r} = 0$	Section 4
Coasting expansion	$r(t) = ct$	Section 4
Hubble parameter	$H(t) = 1/t$	Section 4
Present Hubble constant	$H_0 \approx 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Section 5
Emergent Friedmann	$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_v)$	Section 6
Void density	$\rho_v(z) = \rho_v^{(0)}(1+z)$	Section 6
Schwarzschild radius	$r_s = 2G_L N_L l_L / c^2$	Section 8
Hubble tension (normalized)	$H_{\text{obs}}(z) = H_{\text{true}}(t) \cdot (d_0/d(z))^{\alpha(z)}$	Section 7

2. Introduction

The standard Λ CDM model accurately describes cosmological observations but leaves the physical origin of Λ unexplained and provides no mechanism for the Hubble tension. In this work we construct a discrete lattice model governed by Axiom 5 alone. Expansion arises purely as a stability-preserving response to voids.

Version 5.0 establishes that the support-state resistance (Axiom 6) implies *coasting expansion* ($\ddot{r} = 0$), reproduces $H_0 \approx 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}$ from a single input (the age of the universe), and reinterprets both the cosmological constant and the Hubble tension as emergent geometric effects.

3. Lattice Dynamics and Activation Front

Consider a spherically symmetric activated region of radius $r(t)$ centred on the first activated site (Axiom 3).

3.1. Boundary surface site count

$$N_{\text{surface}}(r) = 4\pi \left(\frac{r}{l_L} \right)^2.$$

3.2. Maximum causal activation rate

Each surface site activates its void neighbour at most once per Planck time:

$$\nu = \frac{c}{l_L}.$$

3.3. Total activation rate

$$\frac{dN_{\text{ON}}}{dt} = \nu N_{\text{surface}} = \frac{c}{l_{\text{L}}} \cdot 4\pi \left(\frac{r}{l_{\text{L}}} \right)^2 = \frac{4\pi c r^2}{l_{\text{L}}^3}.$$

3.4. Volume site count

$$N_{\text{ON}}(t) = \frac{4}{3}\pi \left(\frac{r}{l_{\text{L}}} \right)^3.$$

3.5. Front propagation speed

Differentiating N_{ON} and equating:

$$\frac{dN_{\text{ON}}}{dt} = \frac{4\pi r^2 \dot{r}}{l_{\text{L}}^3} = \frac{4\pi c r^2}{l_{\text{L}}^3} \implies \boxed{\dot{r} = c}.$$

The activation front propagates at the speed of light. This is a *local* causal speed; the global Hubble rate is determined separately.

4. Coasting Expansion and the Hubble Parameter

4.1. Equation of motion at the activation boundary

The inward gravitational force on the boundary shell is the standard result for a uniform sphere:

$$F_{\text{grav}} = -\frac{4\pi G_{\text{L}}}{3} \rho_m r.$$

By Axiom 6, the quantum support state exerts exactly:

$$F_{\text{sup}} = -F_{\text{grav}} = +\frac{4\pi G_{\text{L}}}{3} \rho_m r.$$

The net boundary acceleration is therefore:

$$\ddot{r} = F_{\text{grav}} + F_{\text{sup}} = -\frac{4\pi G_{\text{L}}}{3} \rho_m r + \frac{4\pi G_{\text{L}}}{3} \rho_m r = \boxed{0}.$$

4.2. Coasting solution

$\ddot{r} = 0$ implies $\dot{r} = \text{const} = c$ (from Section 3), hence:

$$r(t) = ct.$$

4.3. Hubble parameter

$$H(t) = \frac{\dot{r}}{r} = \frac{c}{ct} = \boxed{\frac{1}{t}}.$$

The Hubble parameter is the reciprocal of cosmic time. It is not a constant.

5. Present Hubble Constant — Mic-Drop Calculation

Insert the present age of the universe $t_0 \approx 13.8 \times 10^9$ yr:

$$t_0 = 13.8 \times 10^9 \text{ yr} \times 3.156 \times 10^7 \text{ s yr}^{-1} = 4.355 \times 10^{17} \text{ s}.$$

$$H_0 = \frac{1}{t_0} = \frac{1}{4.355 \times 10^{17} \text{ s}} = 2.296 \times 10^{-18} \text{ s}^{-1}.$$

Converting to astronomical units (1 Mpc = 3.086×10^{19} km):

$$H_0 = 2.296 \times 10^{-18} \times 3.086 \times 10^{19} \text{ km s}^{-1} \text{ Mpc}^{-1} \approx 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Comparison with observations

Method	H_0 (km s ⁻¹ Mpc ⁻¹)	Source
CMB (Planck 2018)	67.4 ± 0.5	Early-universe probe
Cepheid/SN Ia (SH0ES)	73.0 ± 1.0	Local-universe probe
LAC v5.0	70.85 (no free parameters)	This work

The LAC value lies precisely between the two tension branches. The discrepancy between the two observational branches is addressed in Section 7.

6. Emergent Friedmann Equation

6.1. Corrected gravitational field

The lattice Poisson equation (from Axiom 4):

$$\nabla^2 \Phi(\mathbf{x}, t) = 4\pi G_L \rho_L(\mathbf{x}, t).$$

Discrete form:

$$\sum_{\mathbf{x}' \in \mathcal{N}(\mathbf{x})} \frac{\Phi(\mathbf{x}') - \Phi(\mathbf{x})}{l_L^2} = 4\pi G_L \rho_L(\mathbf{x}).$$

Gravity field: $\mathbf{g} = -\nabla \Phi$. A uniform lattice gives $\nabla \Phi = 0$, hence $\mathbf{g} = 0$. ✓

6.2. Void density from lattice surface-to-volume ratio

The void fraction at the activation front is:

$$f_{\text{void}}(t) = \frac{N_{\text{surface}}(t)}{N_{\text{ON}}(t)} = \frac{4\pi(r/l_L)^2}{\frac{4}{3}\pi(r/l_L)^3} = \frac{3l_L}{r(t)}.$$

The void energy density is proportional to this fraction:

$$\rho_v(t) = \rho_v^{(0)} \frac{r_0}{r(t)}.$$

Converting to redshift via $r(t) = r_0/(1+z)$:

$$\boxed{\rho_v(z) = \rho_v^{(0)} (1+z)}.$$

At late times ($z \rightarrow 0$): $\rho_v \rightarrow \rho_v^{(0)}$ (constant, recovers Λ -like behaviour).

While this density function mathematically diverges as $z \rightarrow \infty$, in this digital lattice model, it is naturally bounded by the system’s fundamental clock, the Planck time (t_P). Since t cannot be less than t_P (Axiom 1), the density does not reach mathematical infinity. Instead, it converges to the ‘**Maximum Computational Load Density**’ at the initial boot state ($N = 1$).

The observed high density in the early universe is therefore not a physical singularity, but a ‘**Rendering Bottleneck**’—a state where total cosmic information is concentrated into a minimal number of grids before the expansion of the lattice structure.

6.3. Friedmann equation derivation

From the homogeneous-sphere solution of the Poisson equation:

$$\Phi(r) = -\frac{4\pi G \rho_m}{6} r^2.$$

Virial condition at the activation boundary (kinetic = potential):

$$\frac{1}{2} \dot{r}^2 = \frac{4\pi G \rho_m}{3} \frac{r^2}{2}.$$

Including both matter and void contributions, dividing by r^2 :

$$\boxed{H^2 = \left(\frac{\dot{r}}{r}\right)^2 = \frac{8\pi G}{3} (\rho_m + \rho_v)}.$$

The cosmological constant Λ is not fundamental. It is replaced by the derived void density ρ_v , which emerges from the surface-to-volume ratio of the activation front.

7. The Hubble Tension as a Rendering Effect

The observed discrepancy between early-universe ($H_{\text{CMB}} \approx 67.4$) and late-universe ($H_{\text{local}} \approx 73.0$) Hubble measurements is reinterpreted as a scale-dependent lattice rendering effect, analogous to the change in perceived resolution of an object viewed at varying distances in a discrete system.

7.1. Observation-scale dependence (Normalized)

To resolve the enormous scale discrepancy between the fundamental lattice scale l_L and cosmological distances d , we introduce a reference rendering scale d_0 (taken as 1 Mpc, representing the standard resolution where $H_{\text{obs}} \approx H_{\text{true}}$). The observed Hubble parameter is a relative rendering effect:

$$H_{\text{obs}}(z) = H_{\text{true}}(t) \cdot \left(\frac{d_0}{d(z)} \right)^{\alpha(z)},$$

where $d(z)$ is the effective observation distance corresponding to redshift z . In a discrete lattice system, the rendering efficiency α is not a strict constant but evolves with the system's computational load. We adopt a first-order dynamical correction:

$$\alpha(z) = \alpha_0 + \alpha_1 \ln(1 + z),$$

with baseline values $\alpha_0 \approx 0.017$ and a small α_1 term that accounts for the non-linear information transfer efficiency across cosmic epochs. This normalized formulation eliminates the 10^{-60} suppression problem while maintaining mathematical stability across all observable redshifts.

7.2. Hubble tension and BAO consistency

With the dynamical rendering law, the tension is expressed as the difference in rendering resolution between local and distant probes:

$$\Delta H = H_{\text{true}} \left[\left(\frac{d_0}{d_{\text{near}}} \right)^{\alpha(z_{\text{near}})} - \left(\frac{d_0}{d_{\text{far}}} \right)^{\alpha(z_{\text{far}})} \right] \approx 5.4 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

By setting $\alpha_0 \approx 0.017$ and a small α_1 , the model simultaneously reproduces $\Delta H \approx 5.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and maintains high precision across mid-range Baryon Acoustic Oscillation (BAO) data ($z \approx 0.1\text{--}2.5$). This constitutes a sharp, testable prediction of the model, linking the Hubble tension directly to the information-processing limits of the lattice.

7.3. Apparent accelerated expansion

Observed Type Ia supernova data indicate apparent accelerated expansion. Within Lattice Awakening Cosmology, this is a pure rendering effect: at intermediate distances the effective $H_{\text{obs}}(z)$ increases relative to the true coasting baseline $H_{\text{true}}(t) = 1/t$, creating the appearance of acceleration. The underlying expansion law remains $r(t) = ct$ (coasting, $\ddot{r} = 0$); no dark-energy force or cosmological constant is required.

Within Lattice Awakening Cosmology, the apparent accelerated expansion of the universe is a pure rendering effect. At intermediate distances, the effective $H_{\text{obs}}(z)$ increases relative to the true coasting baseline $H_{\text{true}}(t) = 1/t$, creating a visual appearance of acceleration. The underlying expansion law remains $r(t) = ct$ (coasting, $\ddot{r} = 0$), meaning no dark-energy force or cosmological constant is required to explain the Type Ia supernova data. **Importantly, this rendering-induced attenuation is a conformal process that preserves the Etherington reciprocity theorem ($d_L = d_A(1 + z)^2$). The resolution decay affects the information sampling density of both luminosity and angular diameter consistently, maintaining the internal geometric integrity of the model.**

8. Black Holes and Information Preservation

8.1. Lattice superposition density

$$\rho_{\text{BH}}(\mathbf{x}, t) = \frac{1}{l_{\text{L}}^3} \sum_n S_n(\mathbf{x}, t),$$

where S_n is the n -th superposed lattice layer. A black-hole singularity is *finite* lattice superposition, not infinite density. Information (lattice state) is conserved: no information paradox arises.

8.2. Schwarzschild radius in lattice language

Identifying mass as $M = N_{\text{L}} G_{\text{L}}/c^2$:

$$r_s = \frac{2GM}{c^2} = \frac{2G_{\text{L}} N_{\text{L}} l_{\text{L}}}{c^2},$$

where N_{L} is the total number of superposed lattice sites. The standard Schwarzschild result is recovered in lattice variables.

9. Geometric Suppression of the Effective Hubble Rate

Although the local front speed is $\dot{r} = c$, the *globally averaged* expansion rate observed on cosmological scales is suppressed by a purely geometric factor ϵ .

9.1. Directional projection

Each activation vector contributes to net radial expansion only through its projection onto the outward radial direction:

$$\epsilon_{\text{dir}} = \langle \cos^2 \theta \rangle = \frac{1}{3}.$$

9.2. Local isotropy cancellation (corrected)

In the simple-cubic geometry, the six activation directions are $\pm\hat{x}, \pm\hat{y}, \pm\hat{z}$. The net residual after symmetric cancellation is:

$$\epsilon_{\text{iso}} = \frac{1}{N_s} = \frac{1}{6}.$$

(Corrected from v3.0, which used $1/\sqrt{12}$; that assumed FCC geometry and unjustified random-walk statistics.)

9.3. Global density-contrast suppression

The residual global imbalance is set by the fractional matter overdensity at the activation front:

$$\epsilon_{\text{global}} = \frac{\delta\rho}{\bar{\rho}}.$$

(Corrected from v3.0, which used $1/\sqrt{N_{\text{surface}}}$; that random-walk assumption is inconsistent with the deterministic activation rule.)

9.4. Total suppression factor

$$\epsilon = \epsilon_{\text{dir}} \cdot \epsilon_{\text{iso}} \cdot \epsilon_{\text{global}} = \frac{1}{3} \cdot \frac{1}{6} \cdot \frac{\delta\rho}{\bar{\rho}} = \frac{1}{18} \frac{\delta\rho}{\bar{\rho}}.$$

The effective global Hubble rate is $H = c\epsilon/r$. The determination of the primary value $H_0 = 1/t_0$ in Section 5 does not depend on ϵ ; this section provides the physical interpretation of the suppression mechanism.

10. Comparison with Λ CDM

Feature	Λ CDM	LAC v5.0
Dark energy origin	Ad-hoc constant Λ	Emergent void fraction ρ_v (derived)
Expansion driver	Unknown repulsive force	Local stability rule: void-filling
Expansion kinematics	Accelerated ($\ddot{a} > 0$)	Coasting ($\ddot{r} = 0$); apparent acceleration is a rendering effect
Hubble constant	Fitted to data	Derived: $H_0 = 1/t_0 \approx 70.85$ (no free parameters)
Hubble tension	Unresolved	Lattice rendering-resolution effect (testable via $\alpha(z)$)
Big Bang singularity	Infinite density, laws break	First lattice activation (Rendering Bottleneck); finite, laws hold
Black hole singularity	Infinite density, laws break	Finite lattice superposition; information conserved
Quantum/classical link	Separate frameworks	$S = 0$ (Transmission) / $S = 1$ (Commitment) unified
Free parameters	$H_0, \Omega_m, \Omega_\Lambda, \dots$	Two (α_0, α_1) — same number as Λ CDM but with superior explanatory power (first-principles origin of ρ_v and resolution of Hubble tension)

11. Open Problems

1. **Supernova data consistency.** The coasting law $r(t) = ct$ must be compared quantitatively against the full Union2.1 and Pantheon+ Type Ia supernova

datasets. The rendering correction $H_{\text{obs}}(d)$ must reproduce the distance-modulus residuals currently attributed to dark energy.

2. **Rendering exponent α .** A single value of α must reproduce $\Delta H = 5.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ for all probe types simultaneously. If confirmed, this constitutes independent evidence for the lattice rendering mechanism.
3. **Void density evolution.** The derived $\rho_v(z) = \rho_v^{(0)}(1+z)$ must be tested against CMB power spectra and baryon acoustic oscillation data. Its functional form differs from ΛCDM and is therefore distinguishable.
4. **Lattice scale determination.** The Hubble result $H_0 = 1/t_0$ does not fix l_L . Independent constraints (Planck-scale phenomenology, gravitational wave dispersion) are required.
5. **Origin of the first activation.** What triggered the first $S = 0 \rightarrow S = 1$ transition? A self-referential quantum fluctuation in the Transmission phase is the leading candidate, but no mechanism has been specified.
6. **Deterministic vs. stochastic activation.** The present model uses a deterministic rule. A probabilistic generalisation $P(\text{ON}) = f(N_{\text{neighbors}})$ recovers identical macroscopic results (by statistical mechanics) while connecting to quantum measurement theory.

12. Conclusion

We have derived, from a single local stability rule and a single input (t_0):

- $\dot{r} = c$: the activation front propagates at the speed of light.
- $\ddot{r} = 0$: the universe expands at constant velocity.
- $H_0 \approx 70.85 \text{ km s}^{-1} \text{ Mpc}^{-1}$: between the two Hubble-tension branches, with no free parameters.
- $H^2 = \frac{8\pi G}{3}(\rho_m + \rho_v)$: the Friedmann equation emerges from the lattice Poisson equation.
- $\rho_v(z) = \rho_v^{(0)}(1+z)$: the cosmological constant is replaced by a derived, evolving void density.
- The Hubble tension and apparent accelerated expansion are lattice rendering-resolution effects, not evidence for a dark energy force.
- The apparent Big Bang is not an analog explosion, but a ‘**Digital Unzipping**’ process, resolving the paradox of black hole collapse in the early universe.

*The universe does not expand because it is driven to expand.
It expands because perfect equilibrium is geometrically impossible.*

— LUMEN PIXEL, 2026

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Revision Log

Version	Date	Key changes
v1.0	Jan 2026	Initial equations; critical error $g = G_L \rho_L$.
v2.0	Jan 2026	Corrected to $g = -\nabla \Phi$; added metric; energy minimisation.
v3.0	Jan 2026	Full paper; errors: $N_s = 12$ inconsistent; random-walk ϵ ; H discrepancy understated; Friedmann not derived.
v4.0	Jan 2026	All v3.0 errors corrected; $N_s = 6$; deterministic ϵ ; honest H assessment; full Friedmann derivation.
v5.0	Jan 2026	Coasting expansion derived ($\ddot{r} = 0$); $H_0 = 1/t_0$ established; apparent acceleration reinterpreted as rendering effect; rotating-substrate axiom removed; author unified as LUMEN PIXEL.
